

Legume and soy food intake and the incidence of type 2 diabetes in the Shanghai Women's Health Study



Background It has been postulated that a diet high in legumes may be beneficial for the prevention of type 2 diabetes mellitus (type 2 DM). However, data linking type 2 DM risk and legume intake are limited. **Objective** The objective of the study was to examine the association between legume and soy food consumption and self-reported type 2 DM. **Design** The study was conducted in a population-based prospective cohort of middle-aged Chinese women. We followed 64 227 women with no history of type 2 DM, cancer, or cardiovascular disease at study recruitment for an average of 4.6 y. Participants completed in-person interviews that collected information on diabetes risk factors, including dietary intake and physical activity in adulthood. Anthropometric measurements were taken. Dietary intake was assessed with a validated food-frequency questionnaire at the baseline survey and at the first follow-up survey administered 2–3 y after study recruitment. **Results** We observed an inverse association between quintiles of total legume intake and 3 mutually exclusive legume groups (peanuts, soybeans, and other legumes) and type 2 DM incidence. The multivariate-adjusted relative risk of type 2 DM for the upper quintile compared with the lower quintile was 0.62 (95% CI: 0.51, 0.74) for total legumes and 0.53 (95% CI: 0.45, 0.62) for soybeans. The association between soy products (other than soy milk) and soy protein consumption (protein derived from soy beans and their products) with type 2 DM was not significant. **Conclusions** Consumption of legumes, soybeans in particular, was inversely associated with the risk type 2 DM.